

AI-driven surrogate modeling for performance-oriented parametric optimization in sustainable building design: a systematic review[☆]

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ABSTRACT

Performance-oriented optimal design of sustainable buildings is a promising strategy for promoting the development of low-carbon cities. However, traditional physics-based simulation optimization methods require expensive computational resources and time costs. Although Artificial Intelligence (AI) data-driven approaches, as a transformative method, exhibit significant potential in building performance prediction, research on effectively integrating these methods with optimization processes remains relatively limited. This study conducts a comprehensive review of state-of-the-art AI surrogate models for building energy performance prediction and optimization, aiming to identify the most advanced methods or techniques used at each stage of the optimization process. Through a rigorous selection process, 114 relevant studies were selected for detailed critical review. The review meticulously dissects key findings across five dimensions: parametric modeling and performance simulation, dataset construction protocols, state-of-the-art advancements in surrogate modeling, optimization algorithms selection, and multi-criteria decision-making methodologies for design scheme selection. Finally, a technical framework is proposed for AI-driven surrogate modeling applied to multi-objective optimization in sustainable building design. The study further illuminates critical research gaps and recent methodological breakthroughs, while identifying the challenges and opportunities of AI application for future research. For instance, there is limited exploration of deep learning techniques, the interpretability of surrogate models, insufficient optimization with consideration of uncertain parameters, and inadequate discussion on block-scale decarbonization. This review consolidates previous findings and is expected to serve as a valuable resource, providing references guide for the both future academic research and practical implementations of sustainable buildings and blocks resilience design.

1. Introduction

According to the 2025 report released by the International Energy Agency (IEA), global energy demand has experienced a significant surge, exceeding the average growth rate record over the past decade. Notably, the Architectural, Engineering, and Construction (AEC) industry accounts for nearly 60% of the overall increase in electricity consumption within this context [1]. Concurrently, data disclosed by the United Nations Environment Programme (UNEP) indicates that the AEC accounts for 32% of global energy and contributes 34% of greenhouse gas emissions [2]. Given that nearly half of the buildings worldwide will not have been constructed by 2050, adopting effective sustainable measures can cumulatively reduce energy demand by 85%, which is of vital importance for enhancing the performance of buildings throughout

their entire lifecycle [3]. However, the design of sustainable buildings constitutes a complex process, as a multifaceted array of factors influences building performance. To optimize sustainable building design, architects and engineers leverage diverse methodologies to assess the performance of alternative building solutions across these interrelated dimensions.

1.1. Building performance assessment

Building performance can be evaluated through three main methods: Physics-based models (“white box models”), data-driven models (“black box models”), and hybrid methods (“grey box models”) [4–7]. The conventional white-box model entails substantial time consumption and high computational costs [8]. In contrast, data-driven modeling can

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predict building performance rapidly by using historical data to construct input–output models. However, data-driven models typically exhibit lower prediction accuracy compared to physics-based models, and these models often suffer from a lack of interpretability. Finally, hybrid methods combine the interpretability of physics-based models with the efficiency of data-driven models to achieve high-fidelity and rapid prediction capabilities [9]. These grey box models, alternatively referred to as *meta-models* or *surrogate models*, simulate the expensive physics-based models by establishing statistical correlations between the input and output data of the samples, attempting to strike a balance between model performance and interpretability by finding an equilibrium between known and unknown components [10]. Surrogate models accelerate the evaluation of building performance by replacing complex and computationally expensive physics-based engines. Once the surrogate models have been validated to be sufficiently close to the physics-based models within an acceptable error range, they can be employed to rapidly generate high-fidelity predictive data for a given set of specified building information [11].

Considering the complexity of building design and the need for multi-objective optimization (MOO) of building performance, AI assumes a pivotal role in the design process. Machine learning (ML) provides a promising framework for building design, particularly in the context of “black box models” and “grey box models” modeling approaches. Supervised learning algorithms such as support vector machines (SVM), artificial neural networks (ANN), and decision trees have emerged as prominent tools in the field. These algorithms have attracted substantial attention owing to their demonstrated capabilities in interpretability, generalization, and computational efficiency. Surrogate models accelerate the evaluation of building performance by replacing complex physics-based engines [11]. However, surrogate models also have limitations. Trained on databases generated by physics-based simulations, the accuracy of surrogate models relies heavily on the underlying database and the size of the sample data.

1.2. Multi-objective optimization

When integrating surrogate models with optimization algorithms, two primary scenarios arise: (1) hyperparameter optimization (HPO) of the surrogate models, and (2) MOO of building performance. The objective of HPO is to determine an optimal combination of hyperparameters that enables the model to achieve peak performance on validation or test datasets. Most existing studies rely on hyperparameter selection through a trial-and-error approach, where the hyperparameters are chosen based on the empirical insights from prior research. Although this method is labor-intensive and time-consuming, it is effective in developing model accuracy [12]. Several studies have shown that employing advanced optimization techniques (e.g., grid search, Bayesian) can further reduce the prediction error [13,14].

MOO pertains to optimization problems involving multiple conflicting objectives, necessitating a set of solutions that balance these objectives rather than a single optimal solution [15]. When the number of design variables is limited, exhaustive exploration of all possible combinations of variables is feasible. However, the combinatorial space grows exponentially with the increase in variable dimensionality, rendering optimization algorithms indispensable for efficient problem-solving [16]. Commonly used MOO algorithms include: Pareto-dominance-based algorithm [17], decomposition-based model [18], and swarm intelligence algorithm [19]. In addition, sensitivity analysis (SA) helps to screen key design variables for spatial dimensionality reduction and design robustness [20]. As the number of building performance objectives increases, the output of MOO becomes a complex solution set, making it increasingly unfeasible to determine the optimal solution solely by visual inspection. The entropy-based Topsis method is a scientific and effective way to address this issue [21].

Although numerous simulation tools and parametric platforms can identify optimal design solutions via physics-based models, this process

incurs high computational costs and significant time investment. Grounded in data from physical simulations, these AI-driven models enable significantly faster analysis than traditional physics-based methods. While existing literature reviews on AI-driven sustainable building design have explored surrogate models and optimization algorithms employed in optimization processes [4,6,12], they only summarize one or several steps within the optimization workflow, lacking a comprehensive and systematic overview of the entire process. Furthermore, there remains a scarcity of systematic reviews addressing meso-scale building block optimization design. To address these identified gaps, the present review highlights cutting-edge technologies across six sequential components of building-optimization design, systematically integrates them into a comprehensive technical framework, and incorporates mesoscale building-block optimization into its scope. This holistic approach addresses the fragmented coverage in existing studies, provides an integrated methodological reference for both single-building and block-scale sustainable design, and offers actionable guidance for integrating AI technologies with practical architectural and urban planning workflows.

1.3. Scope and main contribution

Driven by global carbon neutrality goals, sustainable buildings, recognized as a key sector for reducing energy consumption and carbon emissions, are progressively integrating deeply with AI technologies. This integration has fostered diverse research directions, ranging from energy efficiency optimization to full-life-cycle management. Surrogate model-based multi-objective optimization has emerged as a critical tool for addressing complex decision-making challenges, as it balances computational efficiency and optimization accuracy. Nonetheless, the AEC industry remains one of the least digitized sectors globally, requiring further efforts to achieve the beneficial adoption of AI technologies. On one hand, the application boundaries, development stages, and potential future evolution paths of AI models in sustainable building scenarios have not yet been systematically organized; on the other hand, there is still a lack of targeted summarization and comparison regarding the combination logic, integration methods, and applicable scenarios of different types of AI models and multi-objective optimization algorithms.

To address these identified gaps, answering the following research questions is critical: (1) What AI-driven surrogate models and multi-objective optimization algorithms are applied in sustainable building studies, and how are they integrated? (2) What challenges arise in the adoption and application of AI-driven surrogate models within the sustainable building field? Thus, a critical examination of surrogate model applications in sustainable buildings is essential to understand trends, barriers, and opportunities. This systematic literature review (SLR) aims to investigate the latest methods of AI-driven surrogate modeling for building parametric optimization under the sustainability paradigm. Notably, this review seeks to synthesize advancements in surrogate modeling-integrated multi-objective optimization, rather than providing a broad overview of all AI applications in building design. Instead, it specifically targets the performance-oriented optimization of discrete building parameters, leveraging AI-driven surrogate models to balance conflicting goals. It does not aim to cover broader architectural design aspects such as spatial organization or social sustainability. The specific objectives of this study are to:

1. critically review existing applications of AI-driven surrogate model technologies in sustainable building parametric design;
2. identify challenges influencing the adoption and practical application of AI-driven surrogate models in the sustainable building industry, and delineate actionable future research directions to address these gaps.

Finally, the remainder of this paper is structured as follows: [Section 2](#)

elaborates on the methodological steps of the systematic literature review. Section 3 illustrates the findings of the published works and the review of the base-building study scale. Section 4 presents a comprehensive review of AI-driven surrogate model modeling process for multi-objective optimization, including surrogate modeling (Section 4.1), sensitivity analysis and uncertainty analysis (Section 4.2), MOO methods (Section 4.3), and decision-making process (Section 4.4). Section 5 proposes a technical framework for future research and discusses the challenges and opportunities in this domain. Finally, Section 6 synthesizes the summary of these findings.

2. Research method

To identify articles pertinent to the AEC industry for review, the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [22] method was employed in the review process to enhance transparency and scientific rigor. The SLR comprises four sequential phases (Fig. 1): Literature Identification, Records Screening, Eligibility Assessment, and Literature Inclusion.

- Literature Identification:** publications were identified through keyword-based searches in the Web of Science database, with a primary focus on data-driven MOO of building energy performance. To refine the search for studies involving ML models and specific building types, additional keywords were incorporated into the search criteria. The final search string comprised (“Multi-objective optimization” OR “Simulation” OR “Building performance” OR “Building design optimization”) AND (“Machine learning” OR “Deep learning” OR “Artificial Intelligence” OR “Meta-model” OR “Surrogate model”) AND (“Building” OR “Block” OR “Commercial” OR “Residential” OR “Hospital”). The initial screening focused on English-language, peer-reviewed journal research articles published between 2015 and 2025. A secondary search was conducted before manuscript submission to mitigate potential omissions of publications emerging between the initial and the secondary search phrase. Studies published within the last decade were selected to ensure the timeliness of the research methods and technologies. Given the research topic of this review, research unrelated to energy performance (e.g., structural reliability design) was excluded. This search strategy yielded 206 publications for subsequent screening.
- Records Screening:** Each paper underwent manual screening based on title, abstract, and keywords, adhering to the following criteria: (1) the research methodology must be AI-driven, (2) the MOO method must be employed, defined as incorporating at least two objective functions, with at least one function addressing building energy performance. Publications utilizing statistical or non-data-driven approaches, as well as studies solely focusing on energy demand forecasting or building environment assessment (without

MOO integration), were excluded. This screening stage resulted in the removal of 71 articles, leaving 135 papers for full-text review.

- Eligibility Assessment:** 22 publications were excluded from the full-text article review due to the absence of case model specifications, incomplete data collection methods, insufficient sample size justification, and optimization methods solely applied to surrogate model performance improvement. In addition, 4 papers were excluded due to incomplete access. The list was expanded by analyzing their references, and another 5 papers were included. This process yielded a final corpus of 114 papers for detailed review (Table A1).
- Literature Inclusion:** A thorough evaluation was carried out to examine the efficacy of AI algorithms in MOO for building performance design, focusing on their applicability, computational efficiency, and optimization outcomes across diverse architectural contexts. Finally, It identifies the research gaps, key challenges, and future directions under different methods.

3. Findings of the systematic literature review

3.1. Publications overview

Fig. 2 provides an analysis of the reviewed literature. Fig. 2(a) shows a steady upward trajectory in research on AI algorithm-based MOO, with a significant acceleration in publications since 2021. As shown in Fig. 2(b), “Energy and Buildings”, “Applied Energy”, and “Journal of Building Engineering” are the leading trio of journals driving scholarly contributions to this field, with Elsevier serving as the primary publisher. Fig. 2(c) classifies case study cities by Köppen-Geiger climate zones, showing that subtropical humid climates (Cfa) and temperate winter dry climates (Cwa) are the most investigated. Fig. 2(d) reveals that residential, office, and school buildings are the most intensively studied building typologies. Fig. 2(e) demonstrates that building envelope design and building geometry are recurrent focal points, primarily owing to their direct impact on energy consumption. However, block morphology remains under-investigated, suggesting a critical need to expand research into holistic urban sustainability frameworks. Most studies address MOO of energy consumption (heating/cooling), thermal comfort, indoor daylighting, and lifecycle costs (Fig. 2(f)), demonstrating a balanced focus on performance, occupant well-being, and economic viability.

We used VOSviewer software to conduct keyword co-occurrence analysis, which is widely used because of its powerful visualization capabilities, efficient co-linear analysis, and dynamic analysis features [23–25]. Fig. 3 depicts the core themes and research priorities in the current AEC industry. The size of spheres represents the frequency of keyword occurrences, while the color indicates the publication time of the studies. It is clear that “multi-objective optimization”, “thermal comfort”, “consumption”, “genetic algorithm”, and “machine learning” are the key themes in this field. By analyzing the time-series trend of the keywords, “machine learning”, “climate change”, “envelope,” and “form” are emerging research topics in the AEC industry. There is a clear connection between the keyword “multi-objective optimization” and others such as “design,” “performance,” and “genetic algorithm,” which indicates that the typical physics-based model for building performance optimization is a common approach. In contrast, “machine learning”, as an emerging research topic, is gradually being applied in the framework of MOO for building performance. Finally, “climate change” and “form” are inadequately addressed in current research, necessitating further exploration into how these factors influence sustainable building performance in future studies.

3.2. Research scale

Before training a surrogate model, constructing a database comprising inputs and outputs is imperative. This process necessitates setting up a base-building model and performing simulations in physics-based models to populate the database. Two types of building research

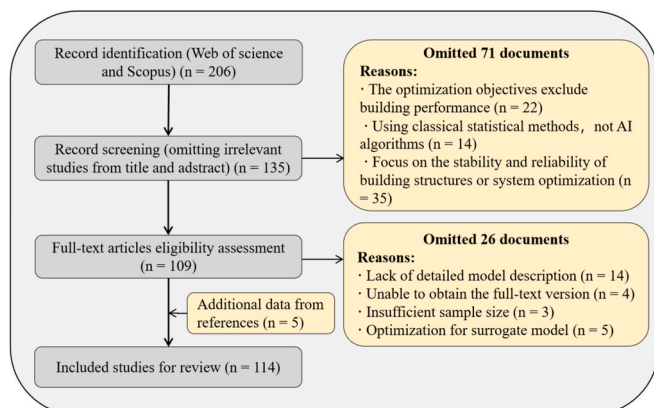


Fig. 1. Literature review process.

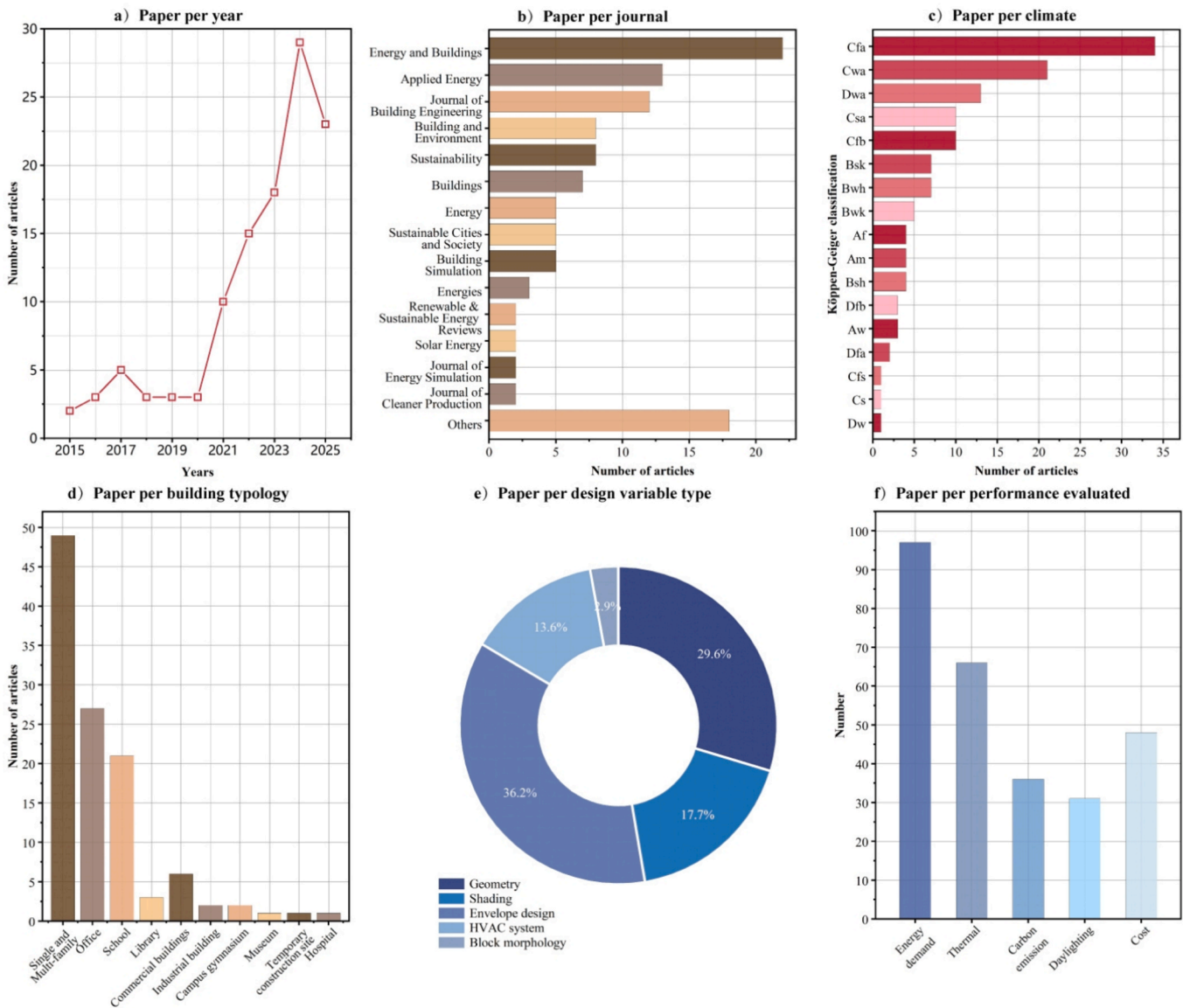


Fig. 2. Publication Overview.

scales are commonly chosen as the base model: 1) a single building within a climate zone or applied to multiple climate zones, 2) covering a building block of multiple building types in a climate zone. Designing for a single climate zone is a deep optimization problem, aiming to derive optimal solutions under specific constraints, while the multi-climate-zones building design emphasizes the model's generalization capability. Building block optimization involves systemic trade-offs. Most studies (82.5%) are centered on a single building within one climate zone, whereas research on a single building applied to multiple climate zones (11.4%) and building block optimization (6.1%) remains underrepresented.

1. Single building: The majority of researchers focus on optimizing the performance of a single building under specific climatic conditions [26–29]. The optimization of a single building for multiple climate zones aims to achieve climate-adaptive design, with a focus on the robustness and generalization capability of the ML model under diverse climatic conditions [30]. Traditional ML methods require separate model training for each climate zone, which is computationally expensive. The transfer learning (TL) framework has recently gained prominence as a highly effective solution to address

this issue. TL achieves knowledge transfer and feature reuse by pretraining-finetuning and multi-task learning, thereby avoiding training the model from scratch [31].

2. Building block: The building block studies require explicit consideration of inter-building interactions (e.g., inter-building effects [32], urban heat island effects [33], and wind interference [34]). Some spatial topological parameters are often used as design variables, such as building layout [35], compactness [36], and development intensity [37]. The optimization workflow typically begins with clustering building typologies, followed by coupling urban microclimate models for integrated energy simulation. Notably, building blocks encounter a combinatorial explosion in design space. Active learning or dimensionality reduction strategies are required to address spatial dependency challenges [38].

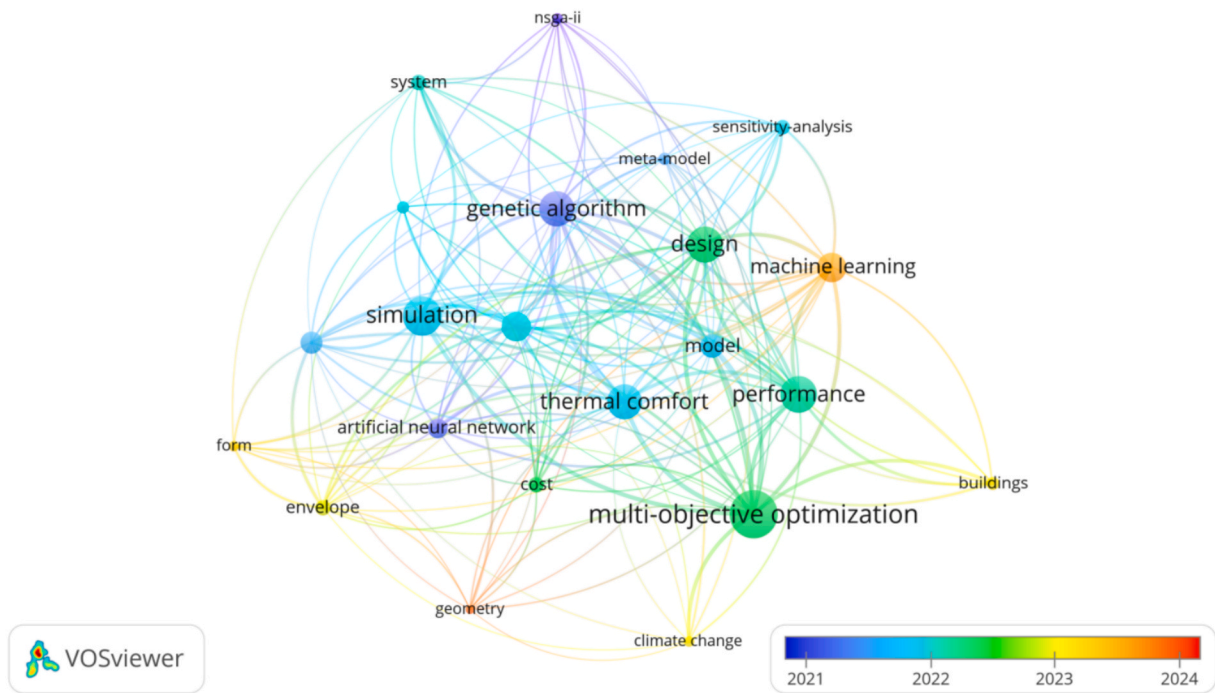


Fig. 3. Co-occurrence analysis map of the included articles.

4. Surrogate modeling for multi-objective optimization

4.1. Surrogate modeling

4.1.1. Model objective functions (output) and design variables (input)

4.1.1.1. Identification of objective function. Objective functions serve as

performance metrics for evaluating design schemes, with their selection guided by building functionality, stakeholder preferences, and building regulations. For academic buildings, daylighting performance indicators are commonly prioritized, such as Useful Daylight Illuminance (UDI) and Daylight Autonomy (DA) [39]. However, residential buildings often adopt thermal comfort metrics, such as Predicted Mean Vote (PMV), and Thermal Comfort Hours (TCH) [40]. In addition, standards such as ISO

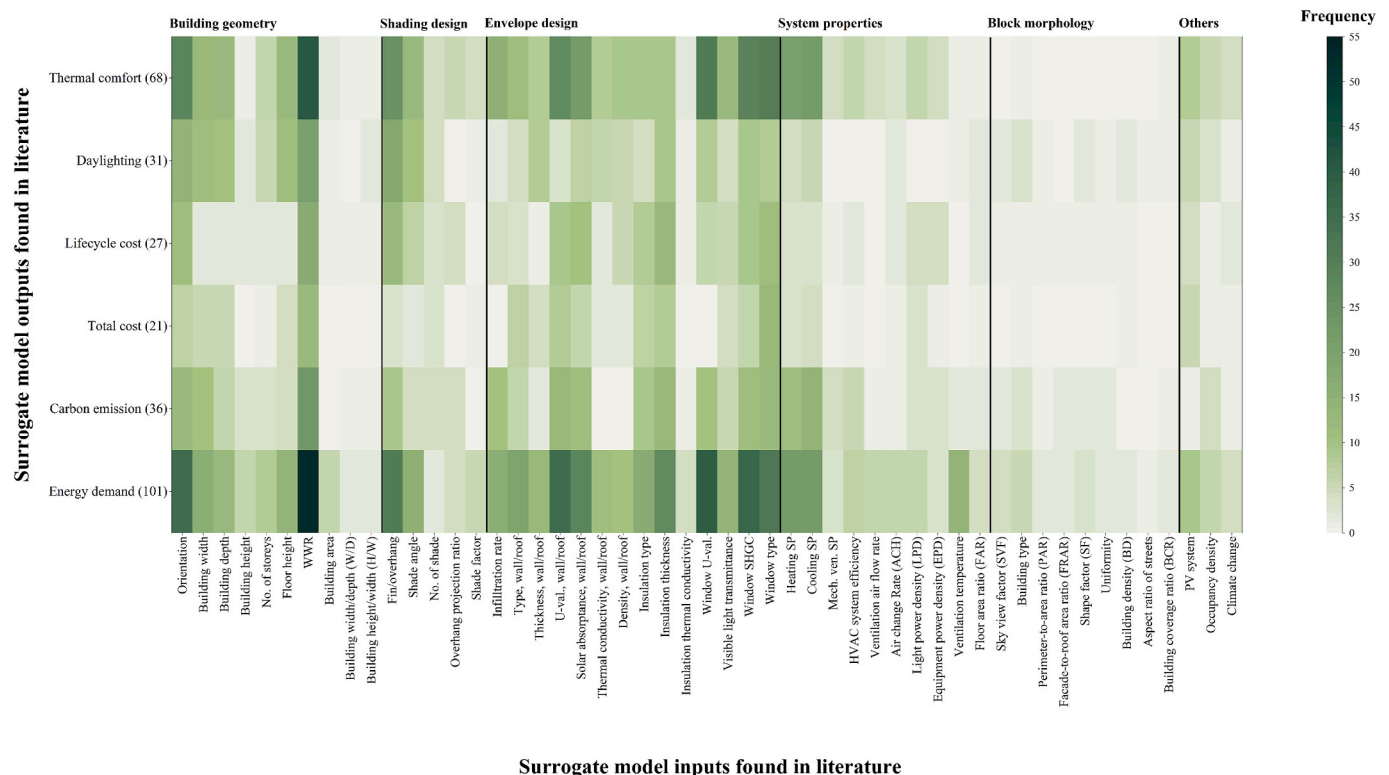


Fig. 4. Usage of objective functions and design variables.

50001, ASHRAE 90.1, and LEED provide technical specifications and evaluation criteria for energy-efficient design. Fig. 4 shows the frequency of specific design variables used for specific different objective functions.

The annual Energy Use Intensity (EUI) was the most common energy target in the optimal design of sustainable buildings. Carbon emission is another critical indicator for evaluating building sustainability. Some studies only consider carbon emissions during the building operation phase. Typically, carbon emission factors are introduced to convert energy consumption into carbon emissions. In addition, some studies still take into account building's life-cycle carbon emissions (LCCE), which encompasses two primary components: operational carbon emissions and embodied carbon emissions (e.g., the production of materials, transportation, construction, and demolition). Total Cost (also known as initial cost) reflects the one-time investment cost, which is contingent on the building area, materials, and usage [41]. In addition, lifecycle cost (LCC) is a critical objective for nearly zero-energy buildings (nZEB) [42,43] and building-integrated photovoltaics (BIPV) buildings [18,44]. Optimizing the BIPV system not only reduces the LCC but also provides long-term additional benefits and returns on investment for building owners. Metrics such as net present value (NPV) [45,46], and payback period [47] are commonly used to evaluate LCC. Additionally, building thermal comfort has also emerged as a key focus of researchers. Key evaluation metrics include: PMV [18,48], Predicted Percentage of Dissatisfied (PPD) [49], TCH [50], Thermal Comfort Time Ratio (TCR) [51,52], and Universal Thermal Climate Index (UTCI) [53]. The building light environment is also an indispensable part of creating a comfortable environment. Objective functions often related to UDI [54,55], sDA [56,57], Spatial Glare Autonomy (sGA) [58]. It reflects researchers' efforts to reduce energy demand while maintaining the applicability of the building environment, transcending traditional energy-first approaches that compromised occupant well-being.

4.1.1.2. Design variables' categories. Building design variables denote adjustable parameters that influence building performance metrics across design, evaluation, and optimization. These variables govern design trade-offs at different project phases.

In terms of a single building, building geometry serves as a core design parameter that defines the structural shape and spatial layout of a building, and its rational design directly influences the building's functional layout and space utilization efficiency. Additionally, as key design variables, shading design and building envelope exert a significant impact on a building's energy consumption level, indoor thermal comfort experience, and overall construction cost by regulating the absorption and dissipation of building heat as well as the penetration path of natural light. Meanwhile, the selection of electrical system and the design of operating parameters further affect building energy consumption and indoor environmental quality, making them indispensable components in the optimal design of a single building. The heatmap reveals that window-to-wall ratio (WWR) [59], orientation [60], fin/overhang [61–64], shade angle [65], solar absorptance [66], U-value [67,68], SHGC [69,70], insulation thickness [71–73], and cooling and heating setpoints [74–76] are the building design variables that have received significant attention from researchers.

Recent scholarly attention has increasingly focused on the optimization design of sustainable building blocks, with BIPV systems emerging as a promising technology for urban-scale sustainability. Zhang et al. [77] suggested that differences in block morphology can increase rooftop power generation by up to 200% under the same planning conditions. Common indicators include building type [78], floor area rate (FAR) [79], sky view factor (SVF) [36], shape factor (SF) [80]. 7 studies explored the relationship between block morphology and building performance at the mesoscale, with six published in the recent 2024–2025. Notably, ten studies compared the sensitivity of optimal building design parameters under future climate change scenarios

versus current conditions, with eight published between 2022 and 2025. This emerging trend highlights growing awareness of integration of BIPV systems within urban block frameworks and climate resilience, though the sample size remains limited, underscoring the need for longitudinal studies on future sustainable building design.

The type and value range of design variables should be determined based on the objective function and comply with building codes, budget constraints, and practical engineering requirements. In addition, the dimension of design variables should be controlled within a reasonable range. An increase in dimensions will lead to an exponential growth in the number of simulation samples required for optimization, significantly consuming computational resources. Although sensitivity analysis can be used for dimensionality reduction, it is still necessary to carefully screen the dimensions of design variables in the early stage of design to reduce redundancy and improve computational efficiency.

4.1.2. Model generation

Currently, various programming language platforms support the implementation of AI algorithms, including Python, MATLAB [81], Grasshopper [82], R [30], Java [83], and Dynamo [84]. Fig. 5(a) indicates that Python and MATLAB dominate as the primary languages for developing surrogate models, owing to their robust ecosystems of machine learning packages and toolboxes. These tools enable designers to implement AI-driven generative design workflows without extensive coding, facilitating the optimization of parametric models for energy performance and spatial geometry.

Selecting an appropriate sampling strategy (also referred to as the design of experiments, DoE) is critical for generating spatial samples. Garud et al. [85] offered a comprehensive overview of the DoE strategy, which can be broadly categorized into static sampling and adaptive (or active) sampling (Fig. 6). In the reviewed literature, static sampling is widely adopted due to its simplicity and high stability. In contrast, dynamic sampling is rarely applied due to issues like high technical complexity, strong dependency, and high costs [86]. Latin Hypercube Sampling (LHS) is the most widely adopted strategy (Fig. 5(b)), as it ensures uniform sample distribution across each parameter's full range while minimizing redundancy [87,88]. This makes it particularly suitable for high-dimensional problems, where it typically requires a sample size of 1.5–10 times the number of input variables [89]. However, the reviewed studies employed much larger sample sizes—up to 30 times the variable count—with 500–3000 samples representing the most common range (Fig. 5(c)).

Yet, it remains unclear which sampling method is the best, as the performance of these techniques depends on specific application scenarios, the number of design variables, and the sample size. Static sampling methods have poor adaptability to imbalanced data, which easily leads to model bias. Additionally, they have low flexibility and cannot cope with dynamic or complex scenarios. Dynamic sampling methods, despite their higher technical difficulty and cost, can dynamically adjust sampling weights based on the model training status and prioritize high-value samples. Compared with static sampling, they are more likely to break through performance bottlenecks and show significant advantages in scenarios with high precision requirements. In terms of the sample size, Xu et al. [47] pointed out that a sample size of more than 1000 is ideal, while the best results are achieved when the sample size reaches 2000. While Etemad et al. [72] found that a dataset consisting of 3,500 samples is the optimal sample size for an accurate surrogate model, while a dataset containing more than 9,900 samples is required to obtain accurate non-dominated solutions for the objective function.

4.1.3. Simulation tools of building performance

After generating a sample space, these samples must be imported into physics-based models for performance simulation. The commonly used base-building modelling tools include: Rhino + Grasshopper, Revit + Dynamo, and SketchUp. These platforms are capable of establishing



Fig. 5. Surrogate model overview.

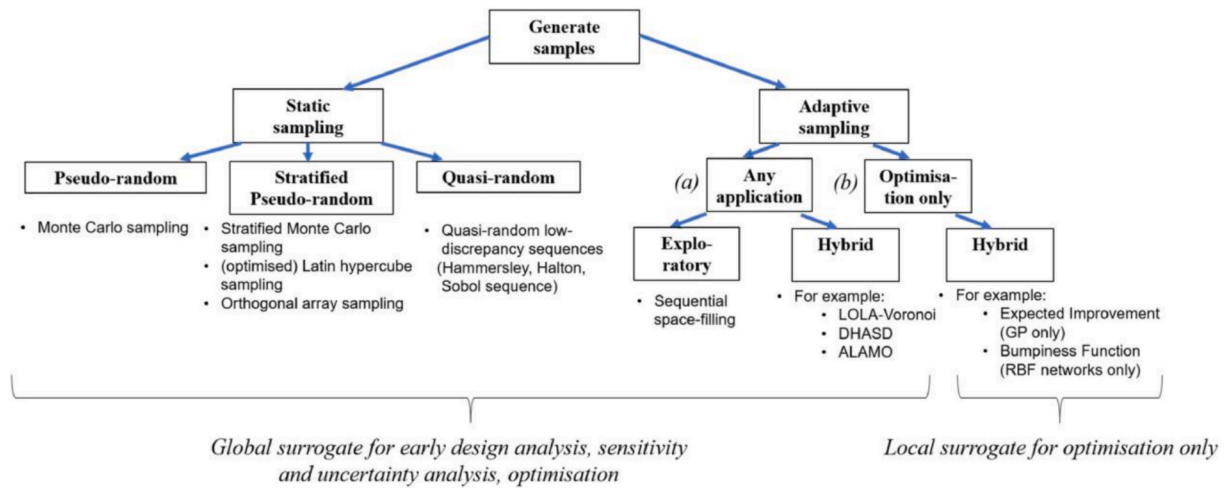


Fig. 6. Overview of sampling methods [36].

detailed and accurate models of building design schemes, with parametric capabilities that facilitate batch modifications of design parameters to generate iterative schemes. Fig. 7 shows the frequency of simulation tool usage in the SLR.

Renowned for its high computational accuracy and extensibility, EnergyPlus serves as the gold standard for both academic research and industry practice. In this review, 95 studies (83.3%) directly used EnergyPlus or other software developed based on EnergyPlus for evaluating building schemes. Radiance is a high-precision light simulation software leveraging Monte Carlo ray tracing to calculate metrics. 31 studies employed Radiance or its derivative tools to assess visual comfort and daylighting performance. In addition, Ladybug & Honeybee provide an integrated platform for built environment performance evaluation. By natively coupling with mainstream simulation engines and supporting parametric design workflows, these tools facilitate real-time feedback between design iterations and performance metrics.

4.1.4. ML prediction model construction

ML prediction model construction consists of four key steps, including data preprocessing, selection of ML algorithms, model training and hyperparameter tuning, and model performance evaluation. While these steps appear mutually independent, they exhibit interactive connections across various stages.

Data preprocessing aims to enhance data quality, consistency, and structural integrity, thereby contributing to improved model generalization capability, and predictive accuracy. Data preprocessing may involve data cleaning, data integration, data normalization, and/or data

reduction to create complete and clean data sets. Data normalization scales features to a predefined range or distribution, mitigating numerical bias and improving model convergence. Common normalization methods include: Min-Max Normalization [90], Z-score Normalization [91], and Robust Scaler Normalization [92]. Dimensionality reduction techniques (e.g., principal component analysis (PCA)) map high-dimensional data onto a lower-dimensional subspace to enhance computational efficiency and mitigate overfitting. The dimensionality reduction and sensitivity analysis for feature selection will be further explored in Section 4.2.

Table 1 presents a synthesis of ML algorithm applications in building performance prediction. Seven major categories of ML algorithms were identified: (1) Neural Network (NN), (2) SVM, (3) Deep Learning (DL), (4) Tree-based Algorithms, (5) Ensemble method, (6) Hybrid Algorithms, and (7) Others. As shown in Fig. 5(d), traditional ML algorithms, such as Artificial Neural Network (ANN), RF, and SVM, have been extensively utilized in building performance optimization. To further pursue the maximization of the potential of single models and enhance the adaptability, two prominent trajectories in ML algorithm development have emerged: (1) the refinement of algorithmic parameters and architectures through metaheuristic optimization, and (2) the integration of disparate ML paradigms to leverage complementary strengths. Although each ML algorithm exhibits unique strengths and limitations, scholars advocate for comprehensive benchmarking across multiple models to identify the optimal algorithm for specific application contexts. Fig. 5(e) shows that 39 studies (34%) conducted comparative analyses of multiple ML models for building performance prediction. Finally, Table A2 lists a catalog of ML algorithms compared in this SLR, accentuating those that demonstrated superior predictive performance and generalization capabilities.

Due to the heavy reliance of traditional machine learning on manually designed features, it exhibits low data utilization and significantly insufficient expressive power when dealing with high-dimensional and non-linear data. Thus, scholars have developed new models to address the limitations of traditional machine learning. By its capability for automatic hierarchical feature learning and higher-order abstraction, DL surpasses traditional machine learning methods in processing unstructured data. Common DL algorithms include: CNN, LSTM, and recurrent neural network (RNN). In addition, traditional single models exhibit weak generalization ability, insufficient robustness, and vulnerability to noise in complex tasks. For instance, decision trees are prone to overfitting, while SVMs have poor adaptability to imbalanced data. Therefore, ensemble models (e.g., Boosting, Bagging, and Stacking [119]) have been developed to address these shortcomings by combining the prediction results of multiple base models, achieving a

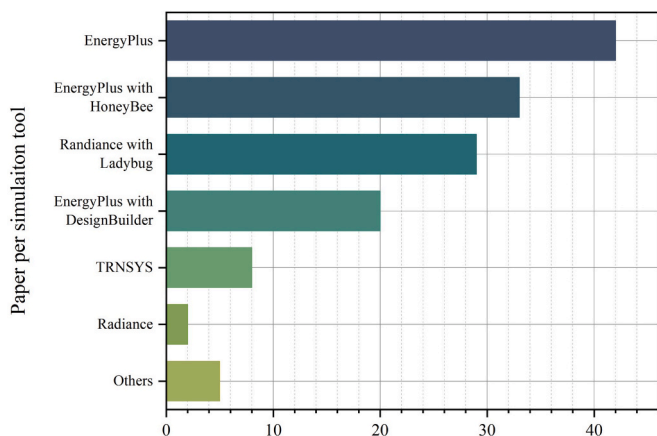


Fig. 7. Frequency of simulation tools usage.

Table 1
List of machine learning algorithms in SLR.

Index	Type	Algorithm	Abbreviation	Reference
1	Neural Network (NN)	Simple Artificial Neural Network	ANN	[92]
2		Multilayer Perceptron Neural Network	MLPNN	[93]
3		Back Propagation Neural Network	BPNN	[94]
4		Radial Basis Function Neural Network	RBFNN	[47]
5		Extreme Learning Machine	ELM	[95]
6		Multilayer feedforward neural network	MFNN	[96]
7		Multilayer Perceptron Neural Network with Artificial Neural Network	ANN-MLPNN	[97]
8		Back Propagation Neural Network with Genetic Algorithm	GA-BPNN	[98]
9		Artificial Neural Network with Least Absolute Shrinkage and Selection Operator	LASSO-ANN	[99]
10		Artificial Neural Network with Principal Component Analysis	PCA-ANN	[100]
11		Artificial Neural Network with Particle Swarm Optimization	PSO-ANN	[101]
12		Back Propagation Neural Network with Particle Swarm Optimization	PSO-BPNN	[102]
13	Support Vector Machine (SVM)	Simple Support Vector Machine	SVM	[103]
14		Support Vector Regression	SVR	[58]
15		Group least square support vector machine	GLSSVM	[84]
16		Least squares support vector machine	LSSVM	[84]
17	Deep Learning	Simple Support Vector Machine with Particle Swarm Optimization	PSO-SVM	[102]
18		Deep Neural Network	DNN	[104]
19		Long Short-Term Memory	LSTM	[17]
20		Multilayer Perceptron	MLP	[95]
21		Convolutional Neural Network	CNN	[33]
22		Deep Belief Network	DBN	[105]

Table 1 (continued)

Index	Type	Algorithm	Abbreviation	Reference
23	Tree-based Algorithms	Graph Neural Network	GNN	[35]
24		Recurrent Neural Network	RNN	[106]
25		Random Forest	RF	[92]
26		Decision tree model	DT	[107]
27		Gradient Boosted Decision Tree	GBDT	[108]
28		Extra Trees Regressor	ETR	[109]
29		Random Forest Regression	RFR	[108]
30		Random Forest with Bayesian Optimization	BO-RF	[110]
31		Classification and Regression Tree	CART	[111]
32		Extremely random forest	ERF	[112]
33		Extremely randomized trees	ET	[38]
34		Fast Random Forest	FRF	[61]
35	Boosting	Gradient Boosted Decision Tree with Particle Swarm Optimization	PSO-GBDT	[102]
36		Random Forest with Particle Swarm Optimization	PSO-RF	[102]
37		Extreme Gradient Boosting	XGBoost	[17]
38		Light Gradient Boosting Machine	LGBM	[113]
39		Gradient Boosting	GBoost	[38]
40		Adaptive Boosting	ADBoost	[112]
41		Light gradient Boosting Machine with Bayesian Optimization	BO-LGBM	[114]
42		Extreme Gradient Boosting with Bayesian Optimization	BO-XGBoost	[69]
43		Categorical Boosting	CatBoost	[60]
44		Categorical boosting + gradient boosting regression tree	CATBoost-GBRT	[13]
45		Gradient Boosting Regressor	GBRoost	[92]
46		Geographically weighted light gradient boosting machine	GW-LGBM	[26]
47	Bagging	Bagging Regressor	BR	[115]
48	Stacking	Support Vector Regression + Random Forest + Extreme Gradient	Stacking	[112]

(continued on next page)

Table 1 (continued)

Index	Type	Algorithm	Abbreviation	Reference
49	Hybrid	Boosting + Gradient Boosted Decision Tree + Adaptive Boosting	ANN-SVM	[84]
		Simple Support Vector Machine with Artificial Neural Network		
		Long Short-Term Memory with Convolutional Neural Network		
50		Simple Support Vector Machine with Convolutional Neural Network	CNN-LSTM	[116]
51		Simple Support Vector Machine with Convolutional Neural Network	CNN-SVM	[33]
52	Gaussian Process	Kriging	Kriging	[80]
53		Gaussian process regression	GPR	[117]
54	kNN	k-Nearest Neighbors	kNN	[118]
55	Transfer learning	Multi-Task Learning	MTL	[86]
56	Bayesian Networks	Bayesian Networks	BNs	[91]

bias-variance balance and improved performance. These models have stronger generalization ability, better robustness, and wide applicability; however, they suffer from high computational costs, poor interpretability, and heavy reliance on the diversity of base models. Other complex algorithms, e.g., k-nearest Neighbors, Gaussian process, and Bayesian networks have also been found in the SLR. Notably, Yuan et al. [50] proposed an automated machine learning (Auto-ML) method, which delivers predictive models with an average accuracy exceeding 95% within a compressed timeframe. This innovation streamlines the model development workflow by automating feature engineering, algorithm selection, and hyperparameter tuning, offering new perspectives for researchers to accelerate AI-driven design processes.

To assess model’s generalizability, a test dataset different from the one used for training is often employed to evaluate the predictive performance on unknown data. Fig. 5(f) shows that splitting the dataset into 80%, and 20% for train, and test respectively, is the most common way [61]. A few studies introduced a validation set, which allows for tuning hyperparameters to avoid overfitting [120]. However, the fixed division of datasets has high randomness, which may lead to a decrease in model generalization ability or distortion of evaluation results. Additionally, it has low data utilization efficiency, resulting in waste of data resources. Therefore, K-fold cross-validation ($k = 5$ [80] or 10 [121]) has been proposed to improve the generalization ability of models and enhance the stability of results, albeit at the cost of additional computational and time resources.

In ML, HPO involves tuning the parameters set prior to model training to improve accuracy. Unlike model weights learned from data, hyperparameters must be specified manually or algorithmically. Table 2 summarizes common HPO methods and their characteristics. Fig. 5(g) and (h) show that 42 studies (37%) employed HPO to identify the optimal configuration for validation or test dataset. The grid search and Bayesian method are widely adopted for their simplicity, wide applicability, and repeatability.

Assessing a model’s reliability and practicality through model performance evaluation is essential. The core of optimization method is to minimize the discrepancy between actual and predicted values, which is typically measured by metrics that quantify error magnitude from diverse perspectives. Key evaluation metrics include the coefficient of determination (R^2), mean squared error (MSE), root mean square error

Table 2

Hyperparameter optimization methods.

Method	Characteristics	Reference
Grid Search	Applicable to a smaller parameter space, global search	[40]
Bayesian	Capable of high-dimensional and complex hyperparameter spaces, intelligent selection of hyperparameter combinations,	[60]
GA	Strong global search ability and adaptability; expensive computation	[95]
Random Search	No requirements for the shape and size of the hyperparameter space; Low precision; unstable results	[13]
PSO	Applicable to continuous parameter optimization; fast convergence; may get trapped in a local optimum	[14]
NSGA-II	Global search; parallel computing; maintaining diversity	[122]
Adam	Applied to deep learning training; may overfitting	[94]
Auto-ML	Automatic model selection and hyperparameter optimization; low explainability	[50]

(RMSE), and mean absolute error (MAE). Supplementary metrics such as mean absolute percentage error (MAPE), coefficient of variation of the root mean square error (cvRMSE), and correlation coefficient (R) are also utilized in specific contexts.

4.1.5. Critical discussion for surrogate modeling

AI-based surrogate models can reduce the time for building performance evaluation [36], the cost-effectiveness of model development is often overlooked. In fact, the processes of dataset acquisition, surrogate model training and application are significantly time-consuming; moreover, with the improvement of computer GPU computing power, the time for full-scale building simulation has gradually decreased. This makes the time advantage of surrogate models less prominent than described in research, and instead, they have to bear additional costs such as model overfitting risks and sample dependence. Furthermore, in small and medium-sized building projects, the prediction accuracy gap between traditional machine learning models (e.g., RF, ANN) and deep learning models is only 3%-8%, but the training time and cost per project of the former are much lower than those of deep learning.

Both actual building operation data and simulated data can be used for dataset construction. Simulated method often simplifies models by fixing some building parameters and ignoring user behavior deviations and equipment aging deviations. This leads to high accuracy of surrogate models in laboratory environments but a significant drop in accuracy when applied to real scenarios. Such research design that strips away real variables cannot meet the needs of the full life cycle of building engineering, and it is essentially aimed at reducing model complexity and improving the efficiency of academic publication rather than solving practical problems. In addition, the ethical issues and acquisition difficulties of real data are generally avoided. Building performance data belongs to the privacy of owners, few studies mention data desensitization measures. The volume of real data is often insufficient, which fails to meet the training requirements of deep learning models and fundamentally undermines the scientific rigor of AI methods.

AI models generally achieved an accuracy of over 85%, yet only 10% provided analyses of the model’s decision-making path (e.g., SHAP method [115]). This tendency of prioritizing predictive accuracy over decision interpretability makes it difficult for designers to judge the rationality of AI-recommended schemes. The “black-box nature” of machine learning models determines their inability to clearly explain the logic behind conclusion generation. This non-interpretability at the technical level essentially arises because AI research in the architectural field overlooks a core characteristic of the architectural design industry. Architectural designers require complete logical traceability, with a clear causal chain from scheme derivation to performance verification. To establish trust among construction practitioners in AI systems, the key lies in enabling them to understand the internal logic of the system’s

decisions. Therefore, it is imperative to introduce interpretable artificial intelligence technologies and construct AI models with decision transparency.

Model generalization constitutes a hidden trap. Although some studies claim to have improved model generalization through transfer learning [91], they deliberately avoid the dual generalization challenge of cross-climate zone and cross-building type. In particular, the transferability of surrogate models across building types represents a major challenge. It is not a simple extension of adjusting building parameters, but involves qualitative differences in building functions, morphological logics, and usage patterns. Existing transfer learning methods are inadequate adapted to this, and studies often use technical packaging to obscure inherent flaw. For instance, Shafiq et al.'s transfer learning framework, though claiming to improve transferability via domain adaptation algorithms and showing high generalization in data from five Indian cities, did not train separately for different building types and is essentially cross-climate zone transfer [29]. Additionally, cross-building type transfer requires re-collection large amounts of real-world data and professional teams to conduct functional adaptability labeling, sharply increasing cost and difficulty. However, studies mostly only emphasize technical advantages while evading discussions on economic costs and scenario adaptability.

There are significant technical barriers to the usability of AI tools. Mainstream AI-based building optimization tools currently rely on programming environments. However, most architects and planners lack the corresponding programming skills, which directly leads to a low practical adoption rate of AI technology in building design. It is evident that the neglect of the designer-friendliness principle in the development of AI tools has become a key barrier hindering the practical application of AI methods.

4.2. Sensitivity analysis and uncertainty analysis

Sensitivity analysis (SA) is employed to evaluate the impact of input parameter variations on outputs, with the primary goal of identifying highly sensitive parameter variables to facilitate dimensionality reduction. SA quantifies the causal relationship between input changes and output fluctuations before surrogate modeling or MOO. This contrasts with uncertainty analysis (UA), which evaluates the probabilistic range of output variations stemming from uncertain inputs [123].

4.2.1. Sensitivity analysis

SA is categorized into (1) Global Sensitivity Analysis (GSA) and (2) Local Sensitivity Analysis (LSA). LSA quantifies parameter sensitivity via partial derivatives or gradients. But its reliance on local linear approximations limits its ability to capture nonlinear interactions or global uncertainty. In contrast, GSA evaluates the impact of input parameter variations across the entire input space, explicitly accounting for nonlinear relationships and inter-variable interactions. As shown in Table 3, commonly used GSA methods include: regression-based methods, variance-based methods, sampling-based, game theory-based, and other methods [6]. Regression-based methods are widely used in SA for their high computational efficiency and ability to provide global explanations. To enhance the effectiveness of SA, Song et al. [122] adopted a combined approach, integrating DMIM and RDB-FAST with regression methods respectively. Although these methods produced consistent results, they lack the corresponding accuracy in practical applications. In addition, Shapley additive explanations (SHAP), a state-of-the-art model interpretability tool, quantifies individual feature contributions to “black-box” ML predictions, offering rigorous explanations of complex models. For instance, Li et al. [115] used the SHAP method to provide feature importance values for each design variable, and revealed the influence mechanism of variables on the urban heat island effect as well as their interaction effects. Despite its utility, SHAP was only employed in 11 studies (10%) of the SLR, highlighting an underexplored opportunity to enhance the transparency of ML

Table 3

Sensitivity analysis methods.

Type	Method/sensitivity indices	Abbreviation	Reference
Regression-based	Partial Correlation Coefficient	PCC	[124]
	Standardized Rank Regression Coefficient	SRRC	[125]
	Spearman correlation coefficient	SPEA	[124]
	Standardized Regression Coefficient	SRC	[90]
	Pearson correlation coefficient	PEAR	[126]
	Partial Rank Correlation Coefficient	PRCC	[71]
Variance-based	Bayesian Treed Gaussian Process	TGP	[59]
	Sobol	Sobol	[117]
	Fourier Amplitude Sensitivity Test	FAST	[30]
	Random Balance Design – Fourier Amplitude Sensitivity Test	RDB-FAST	[122]
Sampling-based	Morris	Morris	[127]
Game theory-based	SHapley Additive Explanations	SHAP	[115]
Other	Delta Moment Independent Measure	DMIM	[122]
	Kolmogorov-Smirnov	KS	[124]
	Fractional Factorial Designs	FFDs	[128]
	Garson	Garson	[129]

applications. This disparity underscores the field's current emphasis on predictive accuracy over interpretability.

4.2.2. Uncertainty analysis

Quantifying output uncertainty, akin to assessing the variability of design variables in the early stage, is often performed in tandem with SA. The quantification methodology relies on sampling strategies to generate representative datasets, with uncertainty analysis (UA) often building upon SA frameworks. As shown in Table 4, the SLR identifies four categories of uncertainty parameters: (1) Weather conditions, (2) Material properties, (3) Occupancy behavior patterns, and (4) Public policy [26].

4.3. Multi-objective optimization methods

4.3.1. Multi-objective optimization algorithms

The surrogate model-based multi-objective optimization method achieves a balance between multiple conflicting performance objectives with lower computational resource and time costs, so as to obtain the optimal set of non-dominated solutions. However, defining optimization problems often involves nonlinear MOO, posing significant challenges in model construction. Researchers have developed diverse AI optimization algorithms inspired by natural processes. GA, Decomposition-based algorithm (DBA), and Population-based algorithm (PBA) are typical representatives of nature-inspired optimization techniques [132]. Table 5 and Fig. 8 summarize the MOO algorithms used in SLR.

GAs employ crossover and mutation operations to explore the

Table 4

Uncertainty Analysis found in SLR.

Index	Type	Parameter	Reference
1	Weather	Future climate change	[130]
2		Thermal conductivity	[131]
3	Material properties	U-value	[103]
4		Window SHGC	[115]
5		Visible light transmittance	[91]
6	Occupancy behavior	Occupancy density	[60]
7		light power density	[67]
8		Equipment power density	[67]
9	Public policy	Discount rate	[99]
10		Electricity price	[79]
11		Equipment energy efficiency	[86]

Table 5
List of Multi-objective optimization algorithms in the SLR.

No	Type	Multi-objective optimization algorithms	Abbreviation	Reference
1	Genetic Algorithm (GA)	Non-dominated Sorting Genetic Algorithm II	NSGA-II	[133,134]
2		Non-dominated Sorting Genetic Algorithm III	NSGA-III	[100]
3		Strength Pareto Evolutionary Algorithm II	SPEA-II	[117]
4	Decomposition-based algorithm (DBA)	Adaptive Weighted Genetic Algorithm	awGA	[107]
5		Multi-Objective Genetic Algorithm	MOGA	[120]
6		Pareto Envelope-based Selection Algorithm II	PESA-II	[117]
7	Particle Swarm Optimization (PSO)	Multi-Objective Evolutionary Algorithm based on Decomposition	MOEA/D	[135]
8		Adaptive Geometry Estimation based Multi-Objective Evolutionary Algorithm	AGE – MOEA	[69]
9		Reference Vector Evolutionary Algorithm	RVEA	[58]
10		Multi-objective selection based hypervolume	SMS – EMOA	[18]
11		Multi-Objective Particle Swarm Optimization	MOPSO	[129]
12		Multi-Objective Grasshopper Optimization Algorithm	MOGOA	[117]
13		Non-dominated Sorting Particle Swarm Optimization	NSPSO	[92]
14		Multi-Objective Water Cycle Algorithm	MOWCA	[117]
15		Multi-Objective Whale Optimization Algorithm	MOWOA	[117]
16		Multi-Objective Salp Swarm Algorithm	MSSA	[117]
17	Others	Multi-Objective Biogeography-based optimization	MOBBO	[95]
18		Multi-Objective Ant Lion Optimize	MOALO	[117]
19		Firefly Algorithm	Firefly	[105]
20		Multi-objective Grey Wolf optimization algorithm	MOGWO	[117]
21		Multi-Objective Manta-Ray Foraging Optimizer	MOMRFO	[121]
22		Two-archive Evolutionary Algorithm for Constrained Multi-Objective Optimization	C-TAEA	[55]
23		Multi-Objective Differential Evolution	MODE	[136]
24		Multi-Objective Dragonfly Algorithm	MODA	[135]
25		Hypervolume Estimation Algorithm	HypE	[82]
26		Adaptive Kriging Surrogate model-assisted Multiobjective Optimization	AKSUMO	[80]
27	Others	Gaussian Process Optimization	GP	[92]
28		Multi-Objective Simulated Annealing	MOSA	[137]
29		Radial-Basis Function Multi-objective Optimizer	RBFMOpt	[92]

Table 5 (continued)

No	Type	Multi-objective optimization algorithms	Abbreviation	Reference
30		Multi-objective Tree-structured Parsen Estimator	TPE	[92]
31		Deep Deterministic Policy Gradient	DDPG	[104]

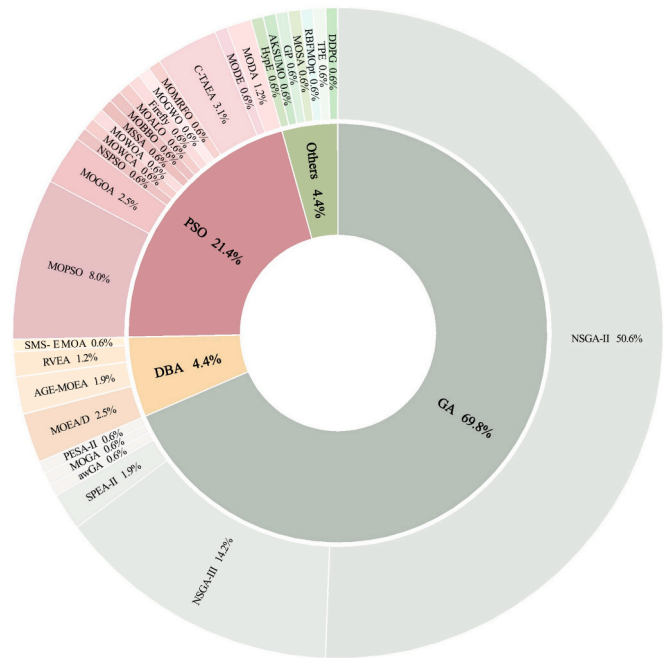


Fig. 8. Frequency of Multi-objective Optimization Algorithms.

solution space globally, intending to identify a set of non-dominated solutions (Pareto optimal set) through non-dominated sorting and crowding distance evaluation of individual solutions. NSGA-II stands as a canonical multi-objective GA, utilized in 50.6% of reviewed studies as the primary optimization technique for its efficiency in balancing solution diversity and convergence. Different from GA, DBA decomposes the multi-objective problem into multiple single-objective subproblems through objective decomposition and inter-subproblem cooperation. By optimizing each subproblem iteratively, DBAs are particularly effective for high-dimensional objective spaces where they can maintain a robust balance between solution diversity and convergence. The MOEA/D is a widely applied DBA technique, valued for its ability to handle complex, multi-dimensional optimization challenges. Unlike GA, which eliminates underperforming individuals, PSO retains all particles in the swarm, allowing even suboptimal solutions to participate in iterations and update their positions based on local and global best values. This characteristic makes PSO particularly suited for continuous optimization problems, where it efficiently navigates smooth solution landscapes. 34 studies employed PSO-based techniques, highlighting their utility in building performance optimization. In addition to these commonly employed optimization algorithms, researchers have explored other alternative techniques to address specific challenges.

Despite the proliferation of these methods, the performance of each algorithm is not consistent across different scenarios, underscoring the critical need for rigorous performance comparisons. 25 studies (22%) systematically evaluated multiple optimization algorithms during the MOO process, using predefined criteria to identify the most suitable approach for their specific context. Finally, Table A3 offers a comparison of the optimization algorithms and highlights the optimization

algorithm that performed better.

4.3.2. Performance evaluation criteria for multi-objective algorithms

Table 6 lists the key metrics used for assessing the performance of MOO algorithms: (1) Efficiency, (2) Diversity, (3) Convergence, and (4) Feasibility Ratio. Among these, the Hypervolume (HV) is a comprehensive metric for evaluating both the quality and diversity of a solution set. It considers not only the convergence (i.e., proximity to the true Pareto front) but also the diversity (i.e., distribution uniformity), making it the preferred choice for researchers.

Xu et al. [83] showed a greater preference for the aNSGA-II, as this algorithm requires less time to reach convergence. On the other hand, Brownlee et al. [75] found that the NSGA-II-Scd algorithm is more superior, as this algorithm can reach a convergent state with fewer iterations. Zorn et al. [92] pointed that RBFMOpt algorithm outperform metaheuristics regarding maximum hypervolume, and the quality of the Pareto fronts. To further improve computing power, researchers have developed a variety of optimization algorithms (e.g., DDPG, MODE, MOWCA, et al.), however, these algorithms have not yet been fully explored in the optimal design of building performance. Finally, Li et al. [136] emphasizes that caution should be exercised when drawing conclusions, as the performance of algorithms varies depending on the selected evaluation indicators. Additionally, since the parameter settings of optimization algorithms have a significant impact on their performance, selecting reasonable optimization algorithm parameters and optimizing these parameters are therefore challenging.

4.3.3. Critical discussion for multi-objective optimization

Pareto-based multi-objective optimization has a core flaw of problem simplification and disconnection from practice. It forcefully compresses the multi-dimensional dynamic system inherent in architectural design into 2–3 quantifiable indicators, while ignoring key but difficult-to-quantify dimensions in practice. Specifically, objectives such as architectural aesthetic constraints, spatial connectivity, accessibility to public service facilities, and traffic noise control are seriously overlooked. This one-sided selection of performance objectives is disconnected from the multi-dimensional needs of real-world design, leading to the difficulty in implementing optimal schemes. Furthermore, even though the Pareto frontier generates multiple non-dominated solutions, only a small number of solutions have value for further evaluation due to hard constraints such as planning regulations and cost budgets in engineering

Table 6
Optimize algorithm performance evaluation metrics.

Type	Metric	Description	Reference
Efficiency	Calculation time	Convergence rate to the Pareto-optimal set	[83]
	Number of iterations	The number of algebraic operations required for the algorithm to reach a stable solution	[75]
Diversity	Hypervolume (HV)	The volume occupied by the solution set within the entire objective function space.	[92]
	Spacing	The standard deviation of the spacing between individuals in the solution set	[107]
Convergence	Generational Distance (GD)	Closeness of a Pareto front approximation to the true or reference Pareto front	[136]
	Inverted Generational Distance (IGD)	Comprehensiveness of the solution set coverage reference frontier	[107]
Feasibility Ratio	Size of Pareto-optimal set	The number of Pareto-optimal solutions in a Pareto-optimal set	[86]
	The number of best Pareto solutions contributed	The number of best solutions contributed by each algorithm to the best Pareto solution set	[96]

practice. The diversity of solutions emphasized by the Pareto method is overstated, and its practical utility is far lower than what research claims.

The high adoption rate of the Pareto method is irrational. Its widespread application stems not from architectural design practice, but from path dependence in academia. Most studies choose it because the multi-objective framework is more conducive to demonstrating academic innovation (e.g., algorithm convergence, solution distribution), which helps meet publication demand. In contrast, the weighted sum method is rejected for being labeled mature method and low innovation. The Pareto method, however, avoids predefined objective weights, thus evading disputes over subjective weight assignment. Additionally, compared with the weighted sum method, the Pareto method incurs significantly higher computational costs from generating non-dominated solutions. This selection logic profoundly reflects the methodological bias in academia of prioritizing academic labels over practical adaptability, which exacerbates the disconnect between AI optimization and design practice. More critically, it deviates from architectural design's essential needs: designers truly require operable schemes with clear priorities, rather than a Pareto frontier that contains numerous invalid solutions.

The generative design of building models has a significant innovation trap. Its algorithmic optimization logic tends to cause scheme convergence, failing to truly capture the diversity that architectural design should have. The Pareto-optimized generative models have a tendency of over-simplification. For example, in generative design at the neighborhood scale, models often only allow adjustments to building height and spacing [38], while ignoring core attributes that affect usability (e.g., public space form, community interaction functions), and excluding dimensions that reflect cultural and aesthetic value (e.g., architectural style coordination, regional materials). This design logic leads to homogeneous neighborhoods and diminishes social attributes. Furthermore, 87% of reviewed studies rely on initial parameter libraries limited to conventional designs paradigms, with the algorithm mutation probability is set below 5%. This makes it difficult for schemes to break through traditional design frameworks. Though Pareto optimization requires multiple solutions, the convergence of the algorithm prioritizes retaining schemes with optimal target values and eliminates schemes with poor target values but innovative design. Eventually, the design diversity of the Pareto frontier is only reflected in parameter fine-tuning, rather than essential innovation.

Furthermore, methodological verification suffers from a distinct lack of closed-loop validation, creating a critical gap between simulated optimization and real-world performance. Specifically, the majority of these studies only verify the effectiveness of AI-optimized schemes using simulated data, while limited study track the actual performance of optimized schemes once constructed. This approach, which validates only simulated outcomes rather than real-world effects, consigns AI optimization to a theoretical exercise.

Existing research also faces the challenge of fragmented interdisciplinary collaboration, marked by an overemphasis on technical disciplines and a lack of integration with design and humanistic perspectives. There is a clear imbalance in the composition of research teams: 89% of studies are led by scholars in computer science and civil engineering, while only 11% include architectural design, sociology, and economics. When developing AI optimization algorithms, computer scientists prioritize algorithm convergence speed over design diversity. Similarly, civil engineering scholars focus heavily on engineering metrics, yet disregard architectural aesthetics and user experience. This gap creates a profound disconnect between AI methods and the tripartite core of architectural design, which integrates environmental, economic, and social considerations.

4.4. Decision-making process

Mathematically, all Pareto optimal solutions are non-dominated and

mutually equivalent in the sense that none can be improved upon without worsening at least one other objective. However, in practical applications, designers must choose a single scheme to inform decision-making. Multiple criteria decision-making (MCDM) techniques are essential for systematically prioritizing and ranking solutions based on predefined design objectives, constraints, and stakeholder preferences.

The decision-making process involves two steps: determining weights for each objective and subsequently applying MCDM techniques. Table 7 lists the methods of weight determination and MCDM techniques identified in the SLR. Subjective Empowerment is usually applicable when the decision maker has rich subjective experience or the targets have clear conditional constraints. For instance, Wang et al. [51] improves the energy efficiency and thermal comfort of buildings through optimization algorithms, and finally selects the optimal Pareto frontier based on the shortest payback period of investment. While Entropy Weight objectively quantifies the importance of indicators via information entropy, rendering it applicable to decision-making contexts with limited prior knowledge. Six MCDM techniques were identified: (1) Simple Additive Weighting method (SAW), (2) Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), (3) VlseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR), (4) Utopia solution (US), (5) Augmented Tchebycheff Scalarizing Function (ASF), and (6) Minimum comfort method.

Table 7
List of Weight determination and MCDM methods.

Step	Method	Description	References
First step: Weight calculation	Entropy Weight	An objective weighting method entirely based on the data itself, avoiding any bias from subjective factors.	[122]
	Uniform Weight	Assigning equal weights to each indicator or determining weights according to a unified rule.	[107]
	Subjective Empowerment	Weighting based on the decision-maker's subjective valuation of the objective functions.	[122]
Second step: MCDM techniques	SAW	Multiply each standard by its corresponding weight and then sum them up to obtain a comprehensive objective function.	[31]
	TOPSIS	Quantify the Euclidean distance (from each solution to the positive negative) ideal solution, and select the solution that is closest to the ideal solution.	[67]
	VIKOR	Reconciling limited decision-making options by minimizing individual regret values and maximizing group utility.	[127]
	US	Emphasizes the pursuit of synergistic optimization of multiple objectives through systematic innovation in decision-making, rather than mere compromise.	[124]
	ASF	Detect the optimal choice solution within a set of possibilities according to various competing criteria.	[130]
	Minimum comfort method	A constraint-based decision-making approach prioritizes the satisfaction of environmental performance indicators (such as thermal comfort and daylighting) before optimizing other objectives.	[74]

In the reviewed literature, Entropy-TOPSIS is a widely adopted decision-making method [55,61]. In addition to the Entropy-TOPSIS, some studies have compared the solutions ranking under different weight configurations using TOPSIS [107]. Several studies evaluated Pareto solutions via Entropy-VIKOR and Entropy-TOPSIS, highlighting the impact of MCDM techniques on outcomes [127]. These methods enable the systematic translation of Pareto sets into actionable design decisions by integrating data-driven objectivity and stakeholder priorities.

4.5. Practical implementations and case studies

While the preceding sections outline the key techniques of AI-driven surrogate models for sustainable building design, their practical value is best demonstrated through practical implementations. This section synthesizes representative case studies that integrate surrogate modeling and MOO, highlighting their application scenarios, key optimization objectives, adopted surrogate algorithms, and tangible outcomes.

At the single-building scale, diverse practical projects validate the efficacy of AI-driven multi-objective optimization. Zhan et al. [99] retrofitted a 7-story senior apartment in Jinan, China, optimizing 20 envelope parameters (e.g., insulation thickness, window-to-wall ratio) to balance thermal comfort, carbon emissions, and cost via the LASSO-ANN surrogate model and C-TAEA algorithm, with the TOPSIS decision-making method employed to select the optimal design scheme. Results indicated that the proposed framework achieved a balanced optimum across conflicting objectives while enhancing computational efficiency. In Singapore's tropical high-density real urban context, Zhang et al. [114] adopted a BO-LGBM surrogate model with SHAP explainability to optimize energy efficiency and indoor thermal comfort of new residential buildings by adjusting residential building geometry and envelope properties, improving the Pareto front distance by over 50% through minor parameter adjustments. Sood et al. [40] applied an XGBoost-NSGA-II framework to semi-detached residences in Dublin, Ireland, reducing computational time by 80% and excessive CO₂ hours by 94% while maintaining comfort by optimizing design parameters (e.g., U-values, window-to-wall ratio, heating setpoint temperature, ventilation rate). Beijing Daxing International Airport, a paradigm of parametric design by Zaha Hadid Architects, showcases sophisticated algorithm-driven optimization for energy efficiency. It utilized genetic algorithms to optimize skylight sunshade layouts, retaining 60% natural light transmission while blocking 40% thermal energy for energy efficiency.

At the building block scale, projects address urban-level sustainability challenges. Shi et al. [138] optimized a residential block in the Yangtze River Delta via Rhino-Grasshopper and NSGA-II, identifying 308 Pareto-optimal solutions that reduced EUI by at least 0.76 kWh/m² and proposed context-specific spatial layout strategies. Shanghai Crystal Plaza encompasses a mixed-use district integrating office, residential, and retail functions [139]. The project adjusted building orientation, geometric form, building type, and spatial layout via parametric tools and genetic algorithms, aiming to meet three key requirements: compliance with residential daylighting standards, adherence to building setback regulations, and high energy efficiency. Ultimately, the project defined the scope of compliant solutions in just two weeks, a process that would otherwise take months with traditional methods. This significantly improved design efficiency and scheme compliance, providing a practical reference for the efficient optimization of mixed-use districts. To address London's housing demand, a high-density residential block used the NSGA-II algorithm to balance FAR (capped at 6.0), daylighting, and energy efficiency [139]. Key parameters include building type and its variants, orientation, spacing, and podium-to-tower ratio. The optimal block design optimizes residential vertical sky component, public space sky exposure, and site coverage, maximizing density while maintaining sunlight standards.

These cases confirm the surrogate modeling's practical utility and reveal key insights: practical projects prioritize stakeholder preferences (e.g., cost, aesthetics) which require flexible MCDM integration; user-friendly platforms (e.g., Grasshopper plugins) enhance interdisciplinary collaboration; and high-quality real-world data is critical for bridging theory and practice. Notably, few projects include post-occupancy evaluation (POE) of AI-optimized schemes, highlighting the need for full-cycle closed-loop verification to refine models and boost real-world applicability.

5. Future trend and direction

5.1. MOO future study framework

Based on the content of the article reviewed in the SLR, this section summarizes the proposed technical framework for future MOO of sustainable buildings (Fig. 9). The framework comprises four modules: (I) Modeling, (II) Simulation, (III) Prediction, and (IV) Optimization. Under these modules, six sequential steps are integrated. The methods or techniques applied in each step are described in detail below.

The technical framework begins with the model and factor setting. This step involves defining building attributes (e.g., research scale, building type, and weather factor), defining design variables, and objective functions. The base model is typically conducted using professional tools, such as Rhino + Grasshopper, Revit, and SketchUp.

This is followed by dataset establishment. The design variables are sampled to form a representative solution space. Common methods include LHS and Monte Carlo sampling, with typical sample sizes from 500 to 3000 samples. Varying sampling methods and sizes are recommended to observe their impact on surrogate model accuracy. The samples are then fed into the performance simulation software to generate the objective function values, creating an input–output database for subsequent analysis. The next step is data preprocessing. The dataset undergoes cleaning, normalization, and SA to further refine the quality and interpretability. GSA methods (e.g., Regression, Sobol's, SHAP) are recommended to identify critical variables and their interactions.

Subsequently, the surrogate modeling process begins. Firstly, data is partitioned into training, validation, and test subsets, or subjected to k-

fold cross-validation (with $k = 5$ or 10) to ensure robust model generalization. ML algorithms (e.g., ANN, tree-based models, deep learning, ensemble methods, or hybrid frameworks) are evaluated based on task-specific requirements. Comparative benchmarking is advised to select the optimal algorithm. During the training and validation process, it is suggested to adjust the hyperparameters of ML algorithm manually or automatically. After that, models are evaluated by key metrics, such as R^2 , RMSE, MSE, and MAE, comparing the deviation of the predicted and simulated values. Once the ML algorithm is validated, the surrogate model serves as a fast proxy for MOO.

Thereupon, the MOO process starts. It is recommended to conduct performance comparisons among multiple optimization algorithms to identify the one that is most suitable for the context. Evaluation metrics, such as efficiency, diversity, convergence, and feasibility ratio, will serve as a metric to evaluate the performance of optimization algorithms. During this process, UA or SA may be conducted to illustrate how uncertainty parameters or design variables shape the Pareto optimal front. This process can determine the priority of design parameters and their range of variation. Scatter plots or parallel coordinate plots are suggested to visualize this process.

Finally, a method of decision-making is required to choose the optimal solution. It requires a two-phase process: first, calculating the weights of the objective function, and second, applying MCDM techniques to select a trade-off scheme, which is based on specific project context. Weight determination methods include Entropy Weight, Uniform Weight, and Subjective Empowerment method. For solution selection, the MCDM techniques are widely used to balance quantitative performance with stakeholder priorities.

5.2. Challenges and future opportunities

The preceding sections have provided a review of the methods and applications of MOO for sustainable buildings using AI algorithms. This section synthesizes the challenges confronting AI algorithms in building design optimization research identified in the SLR, and highlights these challenges as critical future research directions.

(1) Scale expansion & urban-level integration

94% of the studies focus on optimizing individual buildings, but ML models hold untapped potential for larger-scale building block applications. As cities undergo rapid high-density development, urban form and the built environment increasingly influence building energy consumption. Studies have shown that the discrepancy in building energy demand may exhibit a variation factor of up to sixfold under diverse block morphologies [140]. This necessitates a shift from isolated optimization to a holistic framework that accounts for inter-building interactions and coordinates active/passive energy-saving strategies.

(2) Technical advancement & model improvement

The LHS, the dominant static sampling strategy in sustainable building optimization, prioritizes uniform design space coverage but cannot incorporate simulation feedback. This often results in redundant sampling in low-impact regions and insufficient coverage of high-value areas. In contrast, adaptive sampling is a feedback-driven technique that dynamically refines selections based on prior simulation insights. In other fields, adaptive sampling strategies have demonstrated the capability to substantially decrease the number of simulations required for attaining a predefined surrogate model accuracy [141]. Despite this proven efficacy, the AEC industry has yet to undertake a systematic comparison of static and adaptive sampling strategies. Key opportunities for future research may include: efficiency comparison, high-dimensional scalability, and Pareto Front quality. Regarding the significant challenge of model transferability across climate zones and building types, future research needs to make breakthroughs in

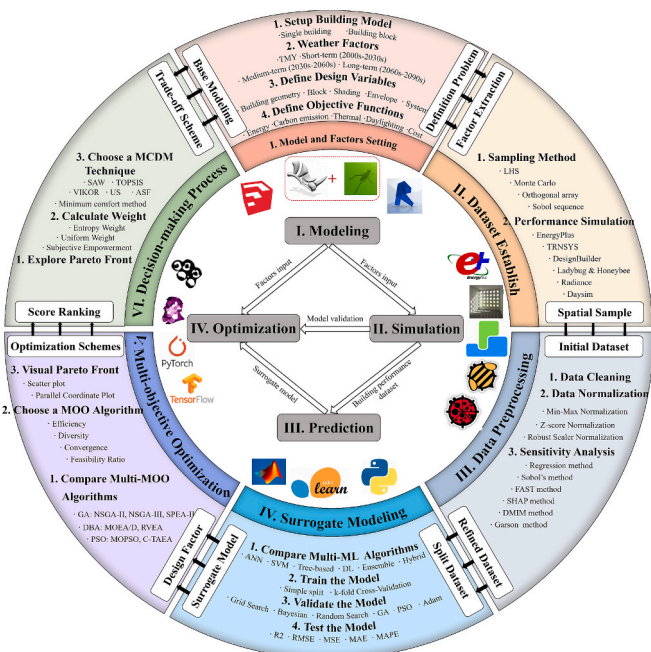


Fig. 9. Framework of AI-driven MOO.

technical adaptation and scenario implementation. A universal feature library should be constructed through multi-modal feature fusion (e.g., building function attributes and user behavior), and an adaptive domain alignment algorithm should be introduced to reduce the impact of cross-scenario data on model accuracy. In addition, it is essential to formulate the effect evaluation standards for cross-type transfer (e.g., error thresholds). Deep learning models excel at automatically identifying complex features and patterns in complex data, demonstrating prowess in tasks like sequential data analysis. In building performance optimization, advanced DL methods (e.g., CNN-LSTM [29], CNN-SVM [33]) have emerged for enhancing prediction accuracy and generalization for complex building systems. However, their applicability and reliability remain underexplored. Key challenges may include scenario-specific validation, interpretability, small sample learning innovation technology, and computational overhead.

(3) Practicality enhancement & industry adaptation

Black-box models are less accepted due to unknown internal structures, making interpretable ML frameworks critical [142]. Techniques such as feature importance analysis and Shapley value-based prediction interpretation bridge data-driven modeling with human-understandable insights, ensuring efficient and actionable design decisions. In addition, Current optimization verification relies mostly on simulations, overlooking real-world variables and leading to performance mismatches post-implementation. Future research should establish full-cycle closed-loop verification to refine models with real operational data. The design logic of optimization algorithms needs innovation to balance technical innovation and engineering practicality. Combining the priority-oriented demand of architectural design, a hybrid optimization framework of the Pareto frontier coupled with dynamic weight matching should be developed to reduce scheme screening costs. For generative design, it is necessary to incorporate non-quantifiable dimensions such as regional culture and architectural aesthetics, and enhance the algorithm's tolerance for unconventional design paradigms. In addition, the value of weighted sum method should be rediscovered, and lightweight optimization tools should be developed for small and medium-sized projects. In the AEC industry, future research directions should include the development of user-friendly plugins and interfaces, designed to streamline their integration into diverse design and engineering workflows.

(4) Resilience & governance strengthening

Given the lengthy lifespan of buildings, addressing the influence of future climate uncertainty on building sustainability during the early design and retrofit stages is critical. This necessitates integrating future weather data to simulate evolving thermal behavior. The future weather data, offered by the Intergovernmental Panel on Climate Change (IPCC), provides General Circulation Models (GCMs) data based on future scenarios through its assessment reports [13]. This technique guarantees that building designs are not only efficient today but also resilient to future environmental and societal shifts, bridging the gap between short-term design decisions and long-term sustainability goals. AI technologies have delivered remarkable outcomes, but inadequate regulation poses significant risks. These include ambiguous accountability in AI-related design accidents, which trigger disputes and enable unfair advantages. Currently, some studies integrate ethics into AI systems [143,144], yet Yampolskiy et al. [145] argue that machines making moral decisions is misleading, and thus proposed a new discipline focused on the safety engineering of artificial intelligence agents. If AI fully replaces human thinking, it will cause catastrophic consequences for social development. However, recognizing this potential risk, governments including the UK have taken action, actively formulating adaptive regulations for AI governance to balance technological development and social security.

6. Conclusion

This review offers a comprehensive analysis of relevant studies using AI-driven surrogate models for multi-objective optimization design of sustainable buildings. Drawing insights from 114 recent studies, it highlights the sequential steps of multi-objective optimization and the methods or techniques employed at each step. In particular, the paper puts forward a technical framework for AI-driven surrogate modeling for performance-oriented parametric optimization in sustainable building design. The framework consists of six key steps: (I) model and factor setting, (II) dataset establishment, (III) dataset preprocessing, (IV) surrogate modeling, (V) multi-objective optimization, and (VI) decision-making process. Finally, we synthesize the challenges identified across the 114 articles and propose actionable opportunities for future sustainable building optimization.

The key conclusions drawn from this systematic literature review are outlined below: most studies focus on optimizing a single building, with limited exploration on building blocks. Envelope design and building geometry are the widely selected variables, while energy demand and thermal comfort are the primary building performance objectives. Researchers widely use Latin Hypercube Sampling to generate datasets, typically ranging from 500 – 3,000 samples. Artificial neural networks, random forest, and support vector machine are the most commonly used machine learning algorithms. In addition, ensemble methods and deep learning methods have been identified for optimization design. To enhance the predictive performance, hyperparameter tuning is commonly performed using grid search and Bayesian optimization, although researchers often neglect this step. NSGA-II is the most widely used method for multi-objective optimization. However, comparative evaluations of multiple optimization algorithms are rare, limiting insights into algorithm-specific strengths across scenarios. The multi-criteria decision-making technique for selecting the trade-off scheme is rarely investigated.

Although AI algorithms have exhibited significant potential in reducing computational costs and expanding the search space, considerable untapped potential remains to be fully harnessed. While the single buildings garnered considerable attention, the energy-saving potential of block-scale form from a mesoscale perspective should not be underestimated. Considering uncertainty parameters (e.g., future climate change) is crucial for enhancing building resilience. To further optimize model accuracy, future research should prioritize adaptive sampling methods and advanced ML algorithms (e.g., transfer learning and deep learning models). The development of user-friendly plugins could narrow the chasm between sophisticated AI algorithms and their real-world implementation.

Overall, AI algorithms have emerged as a transformative technology for sustainable building design. The coupling of physics simulation with AI algorithms significantly enhances the efficiency of the design workflow. This synergy achieves greater economic and environmental benefits at a reduced cost. Amidst the opportunities and challenges presented by this emerging paradigm, architects, engineers, researchers, and policymakers should proactively drive the deep integration of AI algorithms into building design practices, thereby fostering the creation of more sustainable, healthy, and resilient buildings and built environment.

Ethical approval

This study does not contain any studies with human or animal subjects performed by any of the authors.

Declaration of generative AI and AI-assisted technologies in the writing process

While preparing this work, the authors utilized DeepSeek to address grammar issues and enhance readability. Following this, the authors

thoroughly reviewed and edited the content, assuming full responsibility for the publication's content.

CRediT authorship contribution statement

Bo Zhang: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Jingwen Chen:** Visualization, Validation, Software, Methodology, Data curation. **Yumeng Song:** Visualization, Validation, Software, Methodology. **Luyang Zhong:** Visualization, Software, Methodology, Formal analysis. **Lixing Chen:** Software, Methodology, Investigation. **Yingzi Zhang:** Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Data availability

Data will be made available on request.

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